

Phase 2: Requirement Analysis

Software Dependencies

The Smart Lender application is built entirely using Python and its data science ecosystem. The following software components are required to develop, train, and run the project:

- Operating System: Windows 10/11, Linux (Ubuntu), or macOS
- Programming Language: Python 3.10 or above
- Web Framework: Flask 3.1.3 — for building the backend prediction API and serving the web interface
- Machine Learning Libraries: Scikit-Learn 1.7.2 (Logistic Regression, Random Forest), XGBoost 3.2.0 (gradient boosting model)
- Data Handling Libraries: Pandas 2.3.3, NumPy 2.2.6 — for data cleaning, transformation, and analysis
- Model Persistence: Joblib 1.5.3 — for saving and loading the trained model and scaler
- Data Visualization: Matplotlib 3.10.9, Seaborn 0.13.2 — used during exploratory data analysis
- Development & Experimentation: Jupyter Notebook — for dataset exploration and model training/comparison
- Frontend: HTML5, CSS3, JavaScript (Vanilla) — for the web-based user interface
- Version Control: Git and GitHub — for source code management and submission
- Code Editor: Visual Studio Code
- Browser: Google Chrome / Microsoft Edge — to access and test the running web application

Hardware Configurations

As Smart Lender works with a small, structured (tabular) dataset rather than images or deep learning models, the hardware requirements are modest. The recommended minimum configuration is:

- Processor: Dual-core Intel i3 (or equivalent) or higher; Intel i5 recommended for faster model training
- RAM: Minimum 4 GB; 8 GB recommended for smoother performance while training Random Forest and XGBoost models
- Storage: Minimum 500 MB of free disk space for the dataset, trained model files, and project dependencies
- GPU: Not required — the dataset is tabular and small in size, so training runs efficiently on CPU
- Internet Connection: Required for installing Python packages and for pushing/pulling the project via GitHub

Data Specifications

The model is trained on the `loan_approval_dataset.csv` file, which contains historical loan application records. Its structure is summarized below:

- Total Records (rows): 4,269
- Total Features (columns): 13 (12 input features + 1 target variable)
- Target Variable: `loan_status` (binary classification — Approved or Rejected)
- Missing Values: None — the dataset is clean and requires no imputation

Feature Name	Data Type	Description
<code>loan_id</code>	Integer	Unique identifier for each applicant record
<code>no_of_dependents</code>	Integer	Number of dependents of the applicant
<code>education</code>	Categorical	Graduate / Not Graduate
<code>self_employed</code>	Categorical	Yes / No
<code>income_annum</code>	Integer	Annual income of the applicant
<code>loan_amount</code>	Integer	Loan amount requested
<code>loan_term</code>	Integer	Loan repayment term (in years)
<code>cibil_score</code>	Integer	Applicant's credit (CIBIL) score

Feature Name	Data Type	Description
residential_assets_value	Integer	Value of residential assets owned
commercial_assets_value	Integer	Value of commercial assets owned
luxury_assets_value	Integer	Value of luxury assets owned
bank_asset_value	Integer	Value of bank assets/deposits
loan_status (Target)	Categorical	Approved / Rejected — the value to be predicted